ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE E · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice e.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept other correct answers, sensible project decisions and neat, recognisable sketches and charts.

QUESTION 1

1.1 It sets out clearly what the machine must do and the limits it must meet ✓, so that the design can be planned, costed and checked against the requirements when it is finished ✓. (2)

1.2 Any FOUR headings ✓ each with a suitable question ✓, e.g. Function – what must the machine do (feed, punch, eject plates)? Performance – how many plates per hour, how accurate must the holes be? Safety – how will the operator be protected from the punch? Materials – what materials are suitable for the magazine and guards? Cost – what is the budget? Size / environment – how much space is available, is it dusty or oily? Target user – who will operate it and what training do they have? (8)

[10]

QUESTION 2

2.1 Neat freehand sketch ✓ showing: gravity magazine stacked with plates ✓; pusher cylinder under the magazine ✓; sensor at the punching position ✓; punch and chute / bin ✓ (5)

2.2 Original start/stop latch kept ✓; sensor S1 (NO) in series with a CR contact ✓ driving timer T1 ✓; Y1 switched on through a CR / S1 path ✓ in series with an NC contact of T1 ✓ so Y1 switches off after 3 s ✓, e.g. (6)

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POS ─┬──[ Pb1 START ]──┬──[ Pb2 STOP ]──( CR )──────── NEG
     └──[ CR 1.1 ]─────┘
POS ──[ CR 1.2 ]──[ S1 ]──────────────( T1  3 s )──── NEG
POS ──[ CR 1.3 ]──[ S1 ]──[ T1 (NC) ]──( Y1 punch )── NEG
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2.3 Permanent methods cannot be undone without damaging the parts ✓ – e.g. welding, riveting, adhesive bonding ✓. Temporary methods can be taken apart and reused ✓ – e.g. bolts and nuts, screws, circlips, keys and pins ✓. (4)

2.4 Any FOUR ✓✓✓✓: item / part number; description of the part; quantity; material or specification; supplier; unit cost and total cost. (4)

2.5 Capital cost is the once-off cost of buying and building the system (parts, labour, installation) ✓; running cost is the ongoing cost of operating it (electricity, compressed air, maintenance, spares) ✓. (2)

2.6 Inductive sensor symbol (square / diamond with a coil sign) ✓ with three labelled wires: brown +, blue −, black output (PNP) ✓ (2)

2.7 The plates are fed automatically one after another without the operator placing each one ✓, so the machine works faster with fewer stoppages ✓. (2)

[25]

QUESTION 3

3.1 Any FIVE ✓✓✓✓✓: reading technical drawings; measuring and marking out; drilling and tapping; using spanners and torque wrenches; fitting pneumatic tubing and fittings; electrical wiring and crimping; mounting and adjusting sensors; soldering; programming a PLC. (5)

3.2 Chart with the six tasks listed vertically ✓ and days / weeks across the top ✓; bars of sensible lengths in a logical order ✓✓✓ (planning first, testing last, some overlap allowed); total time three weeks ✓; title and labels ✓; neat ✓; milestones / completion shown ✓ (10)

3.3 It identifies the hazards and how serious they are before work starts ✓, so that controls can be put in place to prevent injuries and damage ✓. Any THREE risks ✓✓✓: fingers caught in the punch; flying metal pieces; compressed-air accidents (hoses whipping); electric shock during wiring; heavy lifting; noise. (5)

3.4 A circuit diagram shows how the components are connected electrically and how the circuit works, using symbols ✓✓; a wiring diagram shows the physical layout, the actual position of components, terminals and wire colours / numbers for installation ✓✓. (4)

3.5 Inputs block ✓ (START, STOP, E-stop, plate sensor, cylinder limit switches) → controller (PLC) ✓ → outputs block ✓ (pusher valve, punch valve, indicator lamps, alarm); power supply ✓; arrows showing the direction of signals ✓; labels ✓ (6)

[30]

QUESTION 4

4.1 To check the sequence, timing and sensors safely ✓ without risking damage to the punch, the plates or the operator ✓. (2)

4.2 Any THREE ✓✓✓: all wiring checked against the diagram and terminals tightened; insulation / continuity tests; air fittings tight and pressure regulator set low; guards and E-stops fitted; no tools left in the machine; all people clear. (3)

4.3 Bringing the completed system into service ✓ by testing, adjusting and proving that it works as specified, then handing it over to the user ✓. (2)

4.4 Plates may stop in a slightly different place from the design ✓, so the sensor must be moved to detect the plate reliably at the correct punching position ✓. (2)

4.5 Central topic "Testing the punching machine" ✓ with any FIVE areas ✓✓✓✓✓: emergency stop and guards; magazine feeding; pusher cylinder stroke and speed; sensor detection; punch timing and hole position; ejection into the bin; air leaks; cycle time / output per hour. (6)

[15]

QUESTION 5

5.1 Any SIX sensible items ✓✓✓✓✓✓ with a column to tick and sign, e.g. check guards and E-stop; drain water from the air filter; check air pressure; inspect hoses and fittings for leaks; clean and check sensors; lubricate slides; check punch for wear; check wiring and connectors; check fasteners for tightness. (6)

5.2 With the machine made safe (people clear, punch locked out or air off) ✓, use the PLC software / programming device to monitor or force each output ✓ and check that the correct output LED and device (valve, lamp) responds, then remove all forces ✓. (3)

5.3 Any FOUR ✓✓✓✓: description of the machine and its parts; safety instructions; start-up and shut-down procedures; normal operation; fault-finding / troubleshooting table; maintenance schedule; contact details for support; drawings and spare parts list. (4)

5.4 An unwanted low-resistance path between two points (e.g. positive and negative) so that a very large current flows ✓. (1)

5.5 Any TWO ways with an advantage, ✓✓ each: SMS / e-mail / app notification – reaches the technician anywhere; warning lights and sirens (stack light) – noticed immediately on the floor; dashboard / SCADA alarm screen – shows exactly which fault occurred and logs it. (4)

5.6 A record of faults shows patterns of repeated problems ✓, which helps to plan maintenance, order spares and prevent future breakdowns ✓. (2)

[20]

TOTAL: 100

END OF MARKING GUIDELINE